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INDUSTRIAL MICROBIOLOGY ASSIGNMENT

Case Study on Mushroom
Cultivation Practices at
“Ran bimmal” Mushroom
Cultivation Enterprise,
Badulla, Sri Lanka

Abstract

The cultivation unit mainly produced oyster mushrooms using agricultural waste materials as the growth substrate. Mushroom spawn was obtained from the Horticultural Crop Research and Development Institute (HORDI). The production process included substrate preparation, sterilization, inoculation, incubation, fruiting, harvesting, packaging, and distribution. Observations showed that maintaining suitable environmental conditions, such as temperature, humidity, ventilation, and hygiene, is essential for healthy mushroom growth and high product quality.

The unit had not experienced any major fungal or bacterial contamination. However, preventive measures, including regular sanitation, removal of contaminated logs when necessary, and the use of insect-proof nets, were practiced to reduce contamination risks. The study also revealed that fresh mushrooms have a short shelf life, making same-day marketing important after harvesting.

Waste management practices were environmentally friendly, as spent mushroom substrate was reused as organic fertilizer for vegetable cultivation. Overall, the findings indicate that mushroom cultivation is a sustainable and profitable industry that efficiently utilizes agricultural waste while contributing to food production, income generation, and rural development.

Introduction

Industrial microbiology is a branch of biotechnology that applies microbial sciences to create industrial products in mass quantities. Microorganisms and products are used on a large scale for industrial and commercial purposes. Fungi, bacteria, and yeasts are widely used in industries related to food production, pharmaceuticals, agriculture, environmental management, and biotechnology. Among fungal-based industries, mushroom cultivation has gained noticeable importance due to its nutritional, medicinal, economic, and environmental value.

Mushrooms belong to the phylum Basidiomycota. Mushroom fleshy, macroscopic and spore bearing fruiting structure of a fungus. It can be classified into three categories such as edible, non-edible and poisonous. Edible mushroom is being cultivated since ancient time, however, only 22 out of 2000 species can be consumed. Mushroom cultivation requires relatively little space and these organisms are considered as fast-growing organisms that give the yield within short a time span, hence, mushroom cultivation is a profitable agribusiness for farmers who are interested in an additional income. Global mushroom production exceeds 44 million tons annually. Mushroom possesses high nutritional and medicinal value. Many studies have revealed that the amino acid compositions of mushrooms are similar to animal proteins.

Mushrooms are crucial for sustainable agriculture as they improve soil health, enhance biodiversity, and offer an alternative to conventional farming. They improve nutrient cycling, reduce chemical pesticides, and sequester carbon. Mushrooms also support circular agriculture, and offer economic benefits, especially to marginalized communities. From management waste, soil health, through pest control to food safety, mushrooms have turned the corner in sustainable agriculture because of their minimalistic yet vital impacts on the environmental situation. Their wide use promotes sustainability in agriculture throughout the world. With mushrooms embedded in mainstream agriculture systems, more ecological balance, economic profitability and climate resilience are set

as milestones. Therefore, mushroom cultivation has become an important small-scale industry in many developing countries including Sri Lanka.

This case study was conducted at Ran Bimmal (Golden Mushroom), a small-scale mushroom cultivation enterprise located in Badulla. The industry mainly targets local consumers and contributes to small-scale food production and sustainable agricultural practices in the region. The cultivation unit produces several commercially important edible mushroom varieties including Pink Oyster mushroom (*Pleurotus djamor*), American Oyster mushroom (*Pleurotus ostreatus*), Bhutan mushroom and Abalone mushroom (*Pleurotus cystidiosus*). *Pleurotus sp.* is the second most cultivated edible mushroom worldwide. It has economic and ecological values and medicinal properties. They require a shorter growth time in comparison to other edible mushrooms. Growing oyster mushrooms convert a high percentage of the substrate to fruiting bodies, increasing profitability. They can be cultivated in a simple and cheap way.

The present study was carried out to investigate the industrial microbiological aspects of mushroom cultivation practiced at Ran Bimmal (Golden Mushroom). Special attention was given to the microorganisms involved, raw materials utilized, production technologies, quality control measures, packaging methods and the environmental and economic importance of the industry. Information relevant to the study was collected through field observations, interviews, and literature sources.

Methodology

Direct interviews and structured questionnaire were employed to gather data during the period from 10th to 30th of May 2026 by choosing “The Golden Mushroom” Badulla. The data were collected by visiting the cultivation site and through an online interview. The way the spawn culture is prepared at the laboratory was studied through an online visit to the Horticultural Crop Research and Development Institute (HORDI) - Gannoruwa.

1.Preparation of spawn culture.

1.1. Preparation of PDA media

Table 01-Materials for PDA media.

Ingredient	Amount
Potato	200 g
Dextrose	20 g
Agar	20 g
Distilled water	1 L

Potato dextrose agar media was prepared by using 20g Agar, 20g Dextrose, 200g Potato and 1L. The sliced potatoes were cooked in 500 ml of water for 30 minutes and 20g of agar was mixed simultaneously in 500 ml of water and boiled for 30 minutes. The potato extract was collected after

filtering it and 20g dextrose was added to potato extract. The molten agar was mixed thoroughly with potato-dextrose mixture, and the volume was made up to 1 liter. It was dispensed in sterilized test tubes and bottles. The prepared slant was sterilized in autoclave at 15 pounds pressure and 121°C temperature for 2 hours.

1.2. Raising of Pure Culture

There are two ways of raising pure culture:

Tissue isolation technique- Well grown young mushrooms were selected and cleaned. The mushroom was split lengthwise, and small pieces were taken out from the portion where the gill plate joins the stipe. The tissue was surface sterilized and inoculated PDA media slants. The slants were incubated at 25+1°C for 2-3 weeks. It is more reliable for mushroom culture.

Spore culture technique-

A well-developed fruiting body was selected, and it was cut, and stalk was removed. The mushroom was laid with gills down on glass or typing paper. The spores were collected after 12 hours. It was then inoculated in PDA slants.

1.3. Commercial spawn production

Table 02- Materials for spawn culture.

Ingredient	Amount
CuSO ₄ solution	0.5%
Rice bran	10% (dry weight basis)
CaCO ₃	2% (dry weight basis)
Epsom salt (MgSO ₄)	0.2% (dry weight basis)

The cleaned grains were soaked in 0.5% CuSO₄ for 10 minutes and the soaked grains were thoroughly washed and soaked in tap water for 2 h. After that, the soaked grains were drained, and the excess water was removed. The additives of rice bran at the rate of 10%, CaCO₃ at the rate of 2%, and epsom (MgSO₄) at the rate of 0.2% were added on dry weight basis of the grains and thoroughly and evenly mixed well. The polypropylene bags of 200 gauges and dimensions of 37.5cm length, and 17.5 cm breadth were filled with this grain medium. About 200g of medium was packed in each bag. Then the bags were sealed at the neck region by putting cotton plugs into the conduit/ poly vinyl chloride pipe rings and covered by a piece of aluminum foil by tying a rubber band around the neck. The bags were sterilized in autoclave at 121°C and 15 psi for 30 minutes, and the sterilized bags were allowed to cool for 24 h. Autoclaved bags were shaken well before inoculation. The sterilized bags were kept on the laminar flow. The bags were incubated under aseptic conditions using master spawn as inoculum. 10 to 15 g of master spawn were used for inoculating one bag.

2. Preparation of substrate bags.

Table 03-Materials for substrate mixture.

Ingredient	Amount
Rubber saw dust	20 kg
Rice bran	2 kg
Soy, mung beans and corn flour	1 kg
CaCO ₃	2% (dry weight basis)
Epsom salt (MgSO ₄)	0.2% (dry weight basis)
Water	Required amount for proper moisture content
Medium per bag	800 g

The medium was prepared with the ingredients of Rubber-saw dust, rice bran, soy, mung beans and corn flour and Epsom (MgSO₄) at the rate of 0.2% on the dry weight basis of the substrate. Above mentioned amounts of ingredients were taken and mixed thoroughly with sprinkling water. The mixing process was done using a machine. The correct water content was checked by pressing the medium within the hand. The medium was filled in polypropylene bags of 200 gauges, and the dimensions of 37.5cm length, and 17.5cm breadth. About 800g of medium was filled up to 2/3 of each bag. The bags were then sealed at the neck region by putting cotton plugs into the conduit/ poly vinyl chloride pipe rings and covering a piece of paper by tying a rubber band around the neck. Sealed bags were sterilized by boiling in a steam boiler, and then the sterilized bags were allowed to cool. Those bags were immediately inoculated with spawn prepared earlier. Spawn was inoculated into the sawdust substrate bags under the sterile conditions. One spawn bag was used to fill 60 substrate bags. Bags were kept in dark conditions.

3. Harvesting process.

After the spawn run was complete, the bags were cut open and watered three times a day while supplying light. Fully grown mushrooms were harvested, and the following parameters were recorded: no of days taken for spawn run to complete, no of days taken for first harvest from incubation, no of fruiting bodies per harvest. The harvested mushrooms were packed into sterilized bags and sent to the market.

Results and Observations

Table 1: Summary of observations made at the mushroom cultivation enterprise

Observation	Details
Mushroom species cultivated	American oyster mushroom (<i>Pleurotus ostreatus</i>), Bhutan oyster mushroom (<i>Pleurotus sajor-caju</i>), Pink oyster mushroom (<i>Pleurotus djamor</i>), Abalone mushroom (<i>Pleurotus cystidiosus</i>)
Substrate used	Rubber sawdust mixed with supplementary materials
Spawn source	HORDI (Horticultural Crops Research and Development Institute)
Incubation conditions	Inoculated bags were kept in a dark room until complete mycelial growth was observed
Fruiting conditions	After full mycelial colonization, bags were transferred to light conditions for fruiting and harvesting
Incubation period	Approximately 2-3 weeks
Time required for harvesting	Approximately 1 to 1½ months after inoculation
Hygiene practices observed	Gloves and hair covers were used during handling; only one person entered the cultivation room at a time
Contamination management	Bags contaminated with green mold were completely removed from the cultivation area
Packaging method	Harvested mushrooms were hygienically packed in transparent plastic packs



Figure 1 & 2: Cultivation bags kept in the dark incubation room



Figure 3: Healthy white mycelium growth inside bags



Figure 4: Mushrooms growing from the bags (fruiting stage)



Figure 5: Abalone mushroom (*Pleurotus cystidiosus*)



Figure 6: Pink oyster mushroom (*Pleurotus djamor*)



Figure 7: American oyster mushroom (*Pleurotus ostreatus*)



Figure 8: Bhutan oyster mushroom (*Pleurotus sajor-caju*)



Figure 9 & 10: Packed mushroom products ready for sale



Figure 11: Bag filling machine used for packing mushroom substrate



Figure 12: Mechanical mixer used for blending substrate materials



Figure 13: Steam sterilizer used for sterilizing mushroom cultivation bag



Sprayer

Figure 14: Overhead water spraying system used for maintaining humidity in the mushroom cultivation room

Discussion

Mushroom cultivation is an environmentally sustainable industry that converts low value agricultural wastes into highly nutritious food products. Mushrooms rich in proteins, vitamins, dietary fibers, minerals and bioactive compounds. Therefore, they are important to human nutrients and food security. Mushroom cultivation helps to manage agricultural and industrial waste materials such as sawdust, paddy straw and other lignocellulosic residues. In mushroom cultivation, these waste products convert into edible biomass. This process reduces environmental pollution and promotes waste recycling. Furthermore, mushroom production requires relatively low capital investment, making it suitable for small scale entrepreneurs. The spent substrate can be used as compost after cultivation. Mushroom cultivation also provide economic opportunities through income generation and employment. Due to increasing consumer demand for healthy foods, mushroom production has become an important component of the agricultural and food industries.

The mushroom species cultivated in this industry were pink oyster mushroom (*Pleurotus diamor*), Bhutan oyster mushroom (*Pleurotus sajor-caju*), American oyster mushroom (*Pleurotus ostreatus*), and abalone mushroom (*Pleurotus cystidiosus*). These species belong to the genus *Pleurotus*, which is one of the most widely cultivated groups of edible mushrooms worldwide. These are saprophytic fungi that obtain nutrient by decomposing dead organic matter. They can secrete extracellular enzymes such as cellulases, hemicellulases, and ligninases that breakdown complex lignocellulosic materials present in saw dust and other plant residues into simpler compounds. This enzyme enables *Pleurotus* species to efficiently convert agricultural waste into edible biomass. American oyster mushroom is a one of the most commercially cultivated oyster mushrooms. Because they can rapidly grow and have high nutritional value. Also it can grow on a wide range of substrates. Bhutan is highly adaptable to tropical climatic conditions. Pink oyster mushroom is important for its attractive pink colour, rapid colonize rate. Abalone produces largest fruiting bodies. Also they have longer shelf life compared to others. The vegetative body of these fungi consist of mycelium. The mycelium grows throughout the substrate, absorb nutrient and eventually form fruiting bodies under suitable conditions. Therefore, successful mushroom production depends on maintaining optimal conditions for mycelium growth, substrate colonization and fruiting body development.

In this industry, several steps are followed, including spawn production, substrate preparation, incubation and harvesting. Each step plays a vital role in ensuring successful mushroom cultivation. The scientific principles and microbiological significance of these steps are discussed below.

First, PDA media is prepared. Potato dextrose agar is used for pure culture preparation because it provides ideal nutrient rich environment for fungal growth. Potatoes supply essential nutrients and dextrose act as a readily available carbon source. Agar provides a solid surface for mycelial development. PDA medium is sterilized at high temperature and pressure to eliminate contaminating microorganisms that may compete with mushroom mycelia. Any contaminations at this stage affect the purity and quality of the culture.

Then, Pure cultures are raised either by tissue culture or spore culture techniques. These methods generally used because the resulting mycelium retains the genetic characteristics of the

parent mushroom. Surface sterilization is necessary to remove microorganisms present in mushroom surface. The incubation temperature is 25°C. It provides optimal conditions for mycelial growth. Pure culture preparation is important because it ensure only desired mushroom species is grown without contamination from other fungi or bacteria.

Spawn production is the most important step because the quality of spawn directly affect the mycelial growth, substrate colonization and final yield. Grain spawn production is carried out using grains, rice bran, calcium carbonate and magnesium sulfate. Grains provide large nutrient rich surface area for rapid mycelial growth. Rice bran enhances nutritional value by supplying carbohydrates, proteins, vitamins, and minerals. Calcium carbonate act as buffering agent. It maintains appropriate pH for fungal growth. Magnesium sulfate provide essential minerals for enzyme activity and metabolism. The copper sulfate treatment helps reduce microbial contamination during spawn production.

Substrate preparation is one of the most important stages in mushroom cultivation. Substrate serves as both nutrient source and a physical support for fungal growth. In here rubber saw dust is used as the main substrate component. Sawdust contain cellulose, hemicellulose, and lignin that degrade during enzymatic activity. Nutritional supplements such as rice bran, soybean, flour, mung bean flour, and corn flour is added to improve substrate quality. These materials provide additional carbohydrates, proteins, amino acids, vitamins and minerals. This support vigorous mycelial growth and higher mushroom yield. Calcium carbonate is used to maintain pH and magnesium sulfate is used to supply essential nutrients required for fungal metabolism. Thorough mixing ensures an even distribution of nutrients. Also promote uniform colonization. Proper moisture content is essential for fungal growth. Water is important for enzyme function, nutrient transport and cellular metabolism. Excessive moisture can create anaerobic conditions. Also, it encourages bacterial growth. Insufficient moisture may inhibit mycelial development. Therefore, maintaining optimum moisture content is critical for successful cultivation.

The substrate bags are sterilized using steam to eliminate competing microorganisms. This step is crucial because contaminants such as bacteria and molds can utilize nutrients more rapidly than mushroom mycelia. Also, it suppresses fungal growth. After cooling, immediate inoculation are done under sterile conditions. It minimizes the risk of recontamination. The use of spawn prepared earlier ensures rapid colonization of the substrate. Finally, Bags are kept in a dark condition. It promotes vegetative growth of the mycelium and allowing complete substrate colonization before fruiting begins. Environmental factors such as temperature, moisture, and aeration during incubation affect the rate of mycelial growth and the overall success of mushroom production.

The harvesting stage begins after complete colonization of the substrate by the mycelium. After the completion of mycelium growth bags are cut and water regularly. It helps to maintain the high humidity required for fruiting development. Light exposure stimulates mushroom formation and normal growth. Environmental factors such as humidity, temperature, and ventilation greatly influence mushroom yield and quality. Mushrooms are harvested after full maturation to obtain good quality product and packed in sterilized bags to minimize contamination and maintain freshness. Proper harvesting and handling are important because mushrooms are highly perishable and have a short shelf life.

Since mushrooms are a type of fungi that grow on nutrient rich organic substrates, the same conditions favored by mushrooms can be supportable for the growth of unwanted microbes and pests. So the contamination management and environmental management are two major challenges in mushroom production.

The mushroom cultivation unit that we visited stated that they had not experienced major contaminations during their production process. This indicates the proper hygiene practices, environmental management and quality control measures are effectively maintained in the cultivation. The workers explained that strict cleanliness is maintained throughout the whole process and they only use healthy mushroom spawn from trusted suppliers (HORDI-Gannoruwa). Even though they haven't had any contamination problems, they stated that if any contamination is observed in a substrate bag, the infected one has to be removed and discarded immediately. This practice is important to prevent large scale crop losses because these contaminants can spread quickly to nearby healthy substrate bags via airborne spores, water droplets or physical contact. They told us about several types of commonly reported contaminations and pest infestations in mushroom cultivation industry. As they explained fungal contaminations is one of the most common problems because some fungi compete with mushroom mycelium for nutrients and space. Green mold disease is usually caused by *Trichoderma* sp. which appears as green colored patches on substrate bags. Black mold is usually caused by *Aspergillus niger* which appears as black colored powdery spores. They spread rapidly suppressing the mushroom growth and reducing the yield. poor sterilization of substrates, use of contaminated spawn, excessive moisture and humidity, poor hygiene during inoculation could be some possible causes for fungal contaminations. Bacteria can also be a problem in mushroom cultivation which occurs in excessively wet and nutrient rich conditions. If so the substrate bags would get wet or slimy and produce unpleasant odor. According to them, chlorinated water treatment is one of the best measures to avoid bacterial contaminations. Insect pests also can damage mushrooms and spread contaminants. Common mushroom flies like sciarid flies and phorid flies cause physical damage to mushrooms. Also, they spread fungal spores and bacterial and their larvae feed on mushroom mycelium and fruiting bodies. Mites feed on mushroom mycelium and reduce the growth. The place that we visited use insect proof nets to prevent the entry of insects rather than using a lot of chemical pesticides in order to prevent contaminations.

Mushroom cultivation is highly dependent on controlled environmental conditions like temperature, moisture levels, humidity and ventilation. So when the climate changes suddenly it can affect the growth and productivity of mushrooms. This industry also had experienced harvest reduction during some periods of extremely heavy rainfall and strong sunny conditions compared to normal days. During the periods of heavy rain fall, humidity inside the cultivation house may increase highly and ventilation will become poor. This will support fungal and bacterial contaminations and slowdown the mushroom growth. During extremely hot and sunny conditions, temperature inside the cultivation house may increase. So the substrate moisture may decrease rapidly and mushrooms may dry out faster. It will inhibit mycelial growth and reduce the development of fruiting bodies, lowering the overall yield. So, it is really important to control the growing conditions in order to keep mushroom production stable and to prevent financial losses.

Normally fresh mushrooms have a very short shelf life. This is because of their high moisture content and delicate structure. Therefor mushrooms can get spoiled quickly if not stored properly.

This unit reported that they usually label their mushroom packets with a expiry date of approximately two days from the date of production. This short expiry period ensures that the customers receive fresh and safe products with higher quality and nutritional value. As fresh mushrooms undergo respiration and moisture loss even after harvesting, prolonged storage can lead to conditions like softening, shrinkage, discoloration and unpleasant odor of mushrooms. Therefore, mushrooms must be sold and distributed quickly after harvesting to avoid economic losses. In order to increase the shelf-life mushrooms can be processed in to dried mushrooms, mushroom powder or canned products.

Mushroom cultivation generates relatively low level of waste materials compared to other agricultural industries. The main by-product produced after harvesting is the spent mushroom substrate (what is remaining after the mushroom harvest cycles are completed) along with minor organic residues from processing activities. This may consist of partially degraded sawdust, residual fungal mycelium, remaining organic nutrients and wet organic biomass. This is produced when the substrate doesn't have enough nutrients needed for the mushroom growth and no longer suitable for further production. During the field visit it was observed that instead of discarding the spent mushroom substrate, they reuse it as organic fertilizer for vegetable cultivation. This is an eco-friendly and economically beneficial waste management approach. It will reduce the agricultural waste disposal problems and also minimize the use of chemical fertilizers which may cause several negative effects on the environment as well as the animals. Also, it can be used as animal feed, in vermicomposting and biogas production. Water used for cleaning and spraying also should be managed properly through proper drainage systems in order to prevent microbial contaminations, foul odors and mosquito breeding.

Mushroom cultivation is considered as a highly profitable agro-based industry. This industry requires relatively low initial investment that the other agricultural or livestock industries. It needs low-cost raw materials like saw dust, paddy straw and a small land area. Therefore, small scale farmers also can start this industry with minimal investment. As the production cost is low and market price is good, higher profit margin can be obtained. Mushrooms have a very fast growth cycle. The workers here said that it takes only a few weeks (about a month) from spawning to harvest. Continuous cycles are possible throughout the year. Therefore, faster income generation is an advantage. This industry creates job opportunities for people and provides self-employment opportunities. Even though the unit that we visited is a small-scale industry it has four workers there. As agricultural waste materials are used in the production process this industry adds value to waste materials and reduces disposal costs. So, this is an ecofriendly income source which reduces agricultural waste pollution, produces organic fertilizer as a byproduct and supports integrated farming systems. Nowadays mushrooms have a strong market demand due to their nutritional benefits like high protein content, low fat content and richness in vitamins and minerals. Most of the vegetarian consumers prefer mushrooms as a protein source. Researchers say that edible mushroom usually contain about 2-3 grams of protein per 100 grams on fresh weight of mushrooms.

Conclusion

Mushroom cultivation is an environmental friendly and efficient microbiology based industry. It converts low value agricultural waste materials into a highly nutritious commercially valuable food product. In this industry fungi break down lignocellulosic materials and engage in the role of biodegradation and biomass production. The observed cultivation process depicts the application of industrial microbiology approaches in the whole process. The study showed that successful mushroom cultivation depends greatly on maintaining proper environmental and hygienic conditions throughout the whole process. Factors such as humidity, temperature, sterilization, ventilation, prevention of contaminants and pests are really important for the mycelial growth, fruiting body development, yield of mushrooms and their quality. Therefore, it is crucial to have effective quality control measures and proper pest and disease management in order to obtain a higher productivity and reduce economical losses. Also, mushroom cultivation can be considered as an industry that recycles agricultural waste into edible biomass and organic manure and contributes greatly for the sustainability of the environment. In addition, this industry requires only a small amount of initial investment and creates job opportunities encouraging self-employment. Overall mushroom cultivation can be considered as an economically valuable, sustainable and scientifically important industry that combines microbiology principles with practical industrial applications.

References

- Chand, S. and Singh, B. (2022) 'Mushroom cultivation for increasing income and sustainable development of small and marginal farmers', Asian Journal of Agricultural and Horticultural Research, 9(4), pp. 11–16. Available at: <https://doi.org/10.9734/ajahr/2022/v9i430148> (Accessed: 15 June 2026).
- Department of Agriculture Sri Lanka (n.d.) HORDI Crop – Mushroom. Available at: <https://doa.gov.lk/hordi-crop-mushroom/> (Accessed: 14 June 2026).
- Department of Agriculture Sri Lanka (n.d.) Mushroom Leaflet 1. Available at: https://drive.google.com/file/d/19GP6IK4d_ozIT-n6DoZTjqKhT4IMSFmf/view (Accessed: 13 June 2026).
- Department of Agriculture Sri Lanka (n.d.) Mushroom Leaflet 2. Available at: <https://drive.google.com/file/d/17xiQhSsS3hMpFCBjDE9EsdQ62kvgqjcx/view> (Accessed: 13 June 2026).

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